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Dawini

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List of Abbreviations

- **AI** – Artificial Intelligence
- **NLP** – Natural Language Processing
- **LLM** – Large Language Model
- **GPS** – Global Positioning System
- **UI/UX** – User Interface / User Experience
- **LSTM** – Long Short-Term Memory
- **Bi-LSTM** – Bidirectional Long Short-Term Memory

- **CONV1D** – 1D Convolutional Layer
- **GRU** – Gated Recurrent Unit
- **MSA** – Modern Standard Arabic
- **Q&A** – Questions and Answers
- **TF-IDF** – Term Frequency–Inverse Document Frequency
- **Doc2Vec** – Document to Vector
- **LR** – Logistic Regression
- **RF** – Random Forest
- **MLP** – Multilayer Perceptron
- **SMOTE** – Synthetic Minority Oversampling Technique
- **AHD** – Arabic Healthcare Dataset
- **NLTK** – Natural Language Toolkit
- **PEFT** – Parameter-Efficient Fine-Tuning
- **LoRA** – Low-Rank Adaptation
- **TRL** – Transformer Reinforcement Learning
- **CNN** – Convolutional Neural Network
- **API** – Application Programming Interface
- **VRAM** – Video Random Access Memory
- **STF** – Supervised Fine-Tuning

Abstract

Accessing reliable and culturally relevant medical guidance remains a major challenge for Arabic-speaking individuals. Many users find it difficult to describe their symptoms, understand their condition, or determine the appropriate medical specialist to consult due to the absence of localized, language-specific digital health tools.

Our project introduces Dawini, an AI-powered Arabic medical assistant developed as a mobile chatbot application. Dawini allows users to express their symptoms naturally in Arabic and receive context-aware, AI-generated health suggestions. The system includes a distinct module that recommends the most suitable medical specialty based on structured user-submitted health details using a classification model. A structured database of doctor profiles, including name, specialty, and location coordinates, supports location-aware filtering for nearby specialists. The application also generates and securely stores medical reports in PDF format.

The system is powered by a fine-tuned large language model, Aya Expanse 8B, specifically optimized for Arabic medical understanding. The frontend is built with Flutter for cross-platform compatibility, while the backend utilizes a combination of Flask and FastAPI. Firebase handles user authentication and cloud data storage, and Cloudinary manages the secure storage of medical reports.

By integrating natural language processing, structured triage, and location-aware recommendations, Dawini significantly improves healthcare accessibility for Arabic users, effectively guiding them from symptom description to specialist connection within a simple, intuitive mobile interface.

1. Introduction

Digital health technologies have transformed the way patients search for information, monitor symptoms, and communicate with clinicians. Yet this transformation has been uneven across languages: Arabic speakers—who represent more than 400 million people worldwide—still encounter a shortage of high-quality, language-specific tools for preliminary symptom assessment and provider selection. Most existing platforms either depend on machine-translated content, which can distort medical meaning, or offer rigid symptom checkers that fail to reflect regional terminology and healthcare pathways.

Dawini addresses this gap by integrating three pillars of modern AI-driven healthcare: conversational natural-language understanding, data-driven specialty triage, and location-aware doctor matching. Powered by a fine-tuned large language model (Aya Expanse 8B) and supported by a structured database of physician profiles, Dawini enables users to describe symptoms naturally in Arabic, receive context-appropriate guidance, and connect to nearby specialists—all within a single mobile application.

The system's design deliberately separates free-text symptom dialogue from structured medical triage. Users first engage with an Arabic chatbot to clarify their concerns; they then submit a concise health form that the classifier translates into a recommended specialty. This two-step flow balances conversational flexibility with clinical structure, ensuring both an empathetic user experience and dependable referral accuracy. Supplementary features, such as automated PDF report generation and secure data storage via Firebase and Cloudinary, enhance continuity of care by allowing patients to share their consultation history with physicians.

This report documents the end-to-end development of Dawini, from problem analysis and dataset preparation through model fine-tuning, backend architecture, and user-interface design. It highlights both the technical contributions—such as Arabic-centric NLP pipelines and efficient parameter-efficient fine-tuning strategies—and the practical considerations of deploying an AI service for a linguistically diverse healthcare market.

1.1 Problem definition

Imagine waking up with chest pain, dizziness, or stomach cramps. You open your phone, type your symptoms into a search engine, and are flooded with confusing, often frightening results. For many individuals, this is a daily reality. Without immediate access to a medical professional, people often resort to unreliable online sources, self-diagnose inaccurately, or delay seeking care altogether. This can lead to increased health risks, unnecessary panic, or worse, worsening of a potentially serious condition.

Even when someone decides to visit a doctor, the next struggle begins: **Which specialist should I see?** Many users—especially those with limited medical knowledge—don't know whether their symptoms require a cardiologist, a gastroenterologist, or a general practitioner. Finding the right doctor based on symptoms becomes guesswork.

This challenge is even more significant in the Arab world, where **Arabic-language healthcare platforms are scarce**. Most digital solutions are either in English, rely on rigid symptom checkers, or offer no guidance to nearby, trusted medical professionals. There's a clear gap between what users **need** and what technology **offers**, especially in a culturally and linguistically relevant way.

To address this, our project **Dawini** introduces an AI-powered Arabic chatbot that allows users to describe their symptoms and receive an initial assessment. Doctor specialty recommendations are generated based on user-input health details through a separate classification form. Users can also choose to view nearby

doctors by enabling a location filter. By combining symptom recognition, AI analysis, and optional GPS-based filtering, the system empowers users to make informed medical decisions quickly and confidently.

1.2 Objective

This project aims to develop more than a traditional chatbot; it envisions a reliable and intelligent **digital health companion** tailored for Arabic-speaking users. The system combines natural language processing, medical specialization classification, and doctor matching with real-time recommendations — all embedded within a user-friendly mobile interface. It empowers users to receive preliminary medical guidance in Arabic, directly from their devices, with minimal technical complexity.

The core objectives of the project are as follows:

- **Symptom Interpretation and Conversational Diagnosis**
Design and implement a chatbot capable of understanding user-described symptoms in Arabic using advanced Natural Language Processing (NLP). The system provides context-aware responses and generates medically relevant suggestions. This functionality is powered by the **fine-tuned Aya ExpansE 8B** language model, optimized specifically for Arabic medical dialogue.
- **Medical Specialty Classification**
Rather than recommending doctors through casual conversation, the system includes a dedicated medical form in which users provide structured health information, such as age, weight, height, and symptom descriptions. This data is processed by a **classification model** that predicts the most appropriate **medical specialization** for the case, ensuring that users are matched with doctors suited to their specific condition.
- **Location-Based Doctor Matching**
Leverage GPS data to connect users with nearby doctors within the predicted specialty. Doctor data is managed in a structured database containing detailed profiles, including geographic coordinates, specialty, and contact information. The system can calculate proximity and optionally filter doctors based on distance thresholds to improve accessibility.
- **Medical Report Generation**
Automatically generate a comprehensive medical report in PDF format that summarizes the user's case and the chatbot's insights. These reports are stored using cloud storage solutions and can be reviewed by doctors within the platform. This serves as a shareable, structured summary of the interaction and medical context.
- **Arabic-Centric Interface and Experience**
Develop the frontend using **Flutter**, with a focus on clear UI/UX for Arabic-speaking users. The design prioritizes ease of use for individuals with varying levels of digital literacy, making the system accessible to a broader population.
- **Integration of a Fine-Tuned Arabic Medical LLM**
The project integrates the **Aya ExpansE 8B** language model, which has been specifically fine-tuned for Arabic medical conversation. This ensures more accurate and empathetic chatbot responses, maintaining both medical relevance and linguistic fluency.

By addressing these objectives, the project delivers an intelligent, culturally aware, and highly functional digital healthcare assistant. It bridges the gap between users and medical professionals, enabling better decision-making through AI-assisted guidance, personalized recommendations, and structured communication.

1.3 Motivation

Access to accurate medical advice remains a challenge for millions of Arabic-speaking individuals, especially when faced with sudden health concerns and limited access to trusted professionals. The lack of

culturally and linguistically relevant digital healthcare tools often forces users to rely on vague internet searches or generalized English-language platforms, which can result in confusion, anxiety, and delayed care.

This project is motivated by the urgent need to bridge that gap—to **bring personalized, AI-driven medical guidance directly to users in Arabic**. By combining natural language understanding with real-time GPS-based doctor recommendations, we aim to create a tool that not only understands the user's symptoms but also guides them toward proper medical action.

We are also motivated by the rise of **large language models** and the opportunity they present in creating intelligent, context-aware systems. By integrating a fine-tuned LLM (Aya Expanse 8B) into a fully Arabic interface, we move beyond static chatbots and toward a responsive, conversational healthcare companion.

Furthermore, the decision to use technologies like **Flutter** for a mobile-first experience and to generate **PDF medical reports** stored securely online stems from a desire to make healthcare more accessible, traceable, and user-friendly, even for those with limited technical backgrounds.

Ultimately, **our motivation is to empower users to take control of their health** confidently, quickly, and in their language.

1.4 Challenges and Limitations

Developing Dawini as an AI-powered Arabic medical assistant involved navigating several significant technical and practical hurdles. These challenges not only shaped our development process but also imposed certain limitations on the final system's capabilities and scope, providing clear directions for future work.

- **Scarcity of High-Quality Arabic Medical Datasets:** A primary challenge was the severe lack of publicly available, high-quality Arabic medical datasets suitable for training advanced NLP models. Existing resources, when available, are often unstructured, limited in size, or primarily in English, requiring extensive (and sometimes difficult) translation or domain adaptation. Specifically, obtaining diverse symptom descriptions paired with definitive diagnoses or detailed conversational medical dialogues in various Arabic dialects was particularly difficult. This data scarcity directly impacted the robustness of our model training, potentially limiting the system's ability to handle the full spectrum of medical conditions or diverse user expressions accurately compared to systems trained on richer, language-specific data.
- **Computational Resource Requirements:** Deploying and fine-tuning large language models like Aya Expanse 8B demands substantial computational resources, particularly in terms of GPU memory (VRAM) and processing power. Due to limitations in our available hardware infrastructure, we relied on cloud-based platforms (like Lightning Studio^[16]) for these critical tasks. While enabling the project's completion, this approach introduced dependencies on external services, potential latency in real-time inference, and increased complexity in integration and deployment compared to a fully on-premises solution.
- **Complexity of Arabic Natural Language Processing in Healthcare:** Arabic is a morphologically complex and dialectally diverse language. Users describing symptoms may use colloquial terms, incomplete sentences, misspellings, or even mix Modern Standard Arabic with local dialects. Accurately parsing, understanding the nuances of such informal medical expressions, and generating consistently relevant, medically sound, and empathetic responses posed significant NLP challenges. While Aya Expanse 8B is multilingual, fine-tuning it to reliably interpret the variability of real-world Arabic symptom descriptions and produce highly specific medical guidance required careful data preparation and iterative model refinement.
- **Limitations of the Doctor Database:** The effectiveness of the location-based doctor matching feature is inherently tied to the quality and scope of the underlying doctor database. For this prototype, we

utilized a custom-built database of approximately 400 AI-generated profiles for privacy reasons. This simulated database, while demonstrating the system's functionality, does not reflect the dynamic nature, completeness, or verified accuracy of real-world medical directories. Data points like precise location coordinates, current availability, detailed sub-specialties, or up-to-date contact information for real doctors were not available, limiting the realism and practicality of the current doctor recommendation output.

- **Data Privacy, Security, and Regulatory Compliance:** Handling sensitive user health data (symptoms, medical history summaries, predicted conditions) requires stringent security and privacy measures. While we implemented robust features like Firebase Authentication, minimal data storage, and secure PDF uploads via Cloudinary, achieving full compliance with comprehensive healthcare data regulations (like HIPAA in the US or equivalent regional laws) requires dedicated legal consultation, extensive security audits, and potentially certified infrastructure that were beyond the scope and resources of this academic project. Ensuring patient data confidentiality and integrity remains a critical, complex consideration for any deployment scenario.

2. Related Work

2.1 Introduction and Structure

In recent years, significant efforts have been made to develop chatbots and advance research in Natural Language Processing (NLP), particularly in the domains of healthcare and Arabic language processing. This chapter reviews key contributions in two closely related areas: Arabic medical chatbots and Arabic medical NLP models. By exploring these domains, we aim to highlight various approaches taken to address linguistic challenges in Arabic medical NLP. The objective is to identify common trends, techniques, and limitations that inform the design of our proposed system, **Dawini**—a fine-tuned Arabic medical chatbot powered by a large language model (LLM) and integrated with real-world healthcare functionality.

To provide a clear and logical structure, the chapter is organized as follows:

- **Section 2.2** examines previous research on Arabic medical chatbot systems.
- **Section 2.3** explores Arabic NLP models tailored to the healthcare domain.
- **Section 2.4** presents a comparative analysis, highlighting how our work advances beyond existing approaches in terms of scale, functionality, and integration.

2.2 Arabic Medical Chatbots

Several research studies have been dedicated to the development of Arabic medical chatbots, focusing on text generation, response relevance, and the ability to handle medical conversations. For instance, **Maria Habib et al. (2021)**[1] proposed a predictive text system for generating medical recommendations. Using data from over two million documented consultations on Altibbi, the study focused on five specialties: gynecology, dermatology, psychiatry, urology, and internist diseases. The dataset underwent extensive preprocessing, including tokenization, removal of duplicates, and normalization, with n-gram (3-

gram and 4-gram) representations created for training. The system architecture involved deep learning models, including LSTM, Bi-LSTM, CONV1D, and a hybrid LSTM-CONV1D model, utilizing embeddings like Keras and pre-trained AraVec. These models were optimized for next-word prediction using hyperparameter tuning, dropout, and batch normalization. The CONV1D model achieved the best results across specialties, excelling in accuracy which focused on matching scores.

Similarly, **Abdelhay et al. (2023)**^[2] experimented with three deep learning models LSTM, Bi-LSTM, and Transformers. The dataset was a large Modern Standard Arabic (MSA) healthcare Q&A dataset with over 430,000 questions distributed across 20 medical specializations. They applied word embeddings and conducted data preprocessing for high-quality input. Also, models were designed to improve the accuracy and relevance of responses in medical chatbots. Results demonstrated that the Transformer model outperformed the others, achieving an average cosine similarity of 80.8% and a BLEU score of 58%.

In another notable study, **Boulesnane et al. (2022)**^[3] introduced DZchatbot, a medical assistant chatbot designed for the Algerian Arabic dialect. The authors collected a dataset of 2,150 pairs of medical questions and answers written in the Algerian dialect. The chatbot was developed using three sequence-to-sequence (Seq2Seq) models based on LSTM, Bi-LSTM, and GRU architectures, and evaluated using precision, recall, and F1-score. The GRU outperformed the other models, achieving the best result of 90% accuracy and precision.

2.3 Arabic NLP Models

Beyond chatbot development, several studies have contributed to Arabic medical NLP by creating specialized models and embeddings. **Habib et al. (2021)** developed **AltibbiVec**^[4], a domain-specific word embedding model for the medical and healthcare context in Arabic. the study compared three embedding techniques Word2Vec, fastText, and GloVe.

The models were trained on 1.5 million medical questions in both Modern Standard Arabic (MSA) and Arabic dialects, covering more than 25 medical specialties. The models were evaluated through word clustering, word similarity, and specialty classification using a BiLSTM classifier. Results showed that fastText performed and achieved the highest precision 80.2% and recall 80.1% for classification.

Furthermore, **Faris et al. (2021)**^[5] proposed a multimodal machine learning-based diagnostic system^[5]. This system's dataset was sourced from Altibbi and consisted of more than 246,000 medical consultations, covering over 1,200 diagnoses and 7,300 symptoms. The approach utilized TF-IDF, hashing vectorizer, and Doc2Vec embeddings as feature extraction techniques, as well as machine learning classifiers such as Logistic Regression (LR), Random Forest (RF), and Multilayer Perceptron(MLP). The fusion of the two modalities—structured symptoms and free-text questions—enhanced diagnostic accuracy, achieving a final 84.9% accuracy using a fusion-based multiplication method.

Finally, **Alomari et al. (2023)**^[6] presented a machine learning-based approach to specialty detection. The dataset comprises over 523,000 medical consultations spanning 45 specialties with a highly imbalanced distribution. To address the challenge of class imbalance, the study utilized oversampling techniques like the Synthetic Minority Oversampling Technique (SMOTE) and incorporated class

reweighting and keyword identification. The model architecture was primarily BiLSTM networks with various configurations of nodes (15, 30, 64, 128), complemented by fine-tuned embeddings such as AraVec, AraBERT, and custom-trained AltibbiBert for semantic understanding. Preprocessing involved removing noise and employing TF-IDF and word embeddings to represent textual data. The best results were achieved by combining SMOTE with keyword-based reweighting (factor 15), improved recall, and F1-score for rare classes.

As part of our system's specialty classification module, we adopted the **AltibbiVec**[4] word embedding model to enhance the representation of Arabic medical terms. Specifically, we utilized the fastText-CBOW-100 variant, selected based on its strong performance in prior evaluations by **Habib et al. (2021)**. By embedding user symptom inputs using this model, we improved the semantic understanding of medical expressions, allowing for more accurate mapping between symptom descriptions and relevant specialties. This integration significantly contributed to the classifier's precision and relevance within the Dawini system

2.4 Project Contribution and Novelty

While previous research has made significant strides in developing Arabic medical chatbots and NLP models, our project—**Dawini**—introduces key distinctions in both methodology and practical scope.

2.41 Model Scale

In contrast to earlier systems that rely on classical machine learning or deep learning models such as LSTM, Bi-LSTM, or Transformers (e.g., Abdelhay et al., 2023; Boulesnane et al., 2022), our chatbot leverages a **fine-tuned large language model (LLM)**—**Aya Expansive 8B**[8]. This allows for a deeper contextual understanding and more coherent, accurate responses tailored to Arabic-speaking users.

2.4.2 Dual-Interface Healthcare Integration

Dawini provides two main user interfaces:

- **Chatbot Interface:** Allows users to describe symptoms in Arabic and receive informed, multi-turn responses from the LLM, improving on static Q&A formats used in earlier systems.
- **Classifier Form Interface:** Collects structured personal and health data to recommend a relevant medical specialty, supporting basic triage and connecting users with appropriate care.

This dual approach moves beyond traditional chatbot designs by offering both conversational support and actionable medical direction. Detailed system architecture and implementation aspects are discussed in the next chapter.

3. Proposed Model System Specification

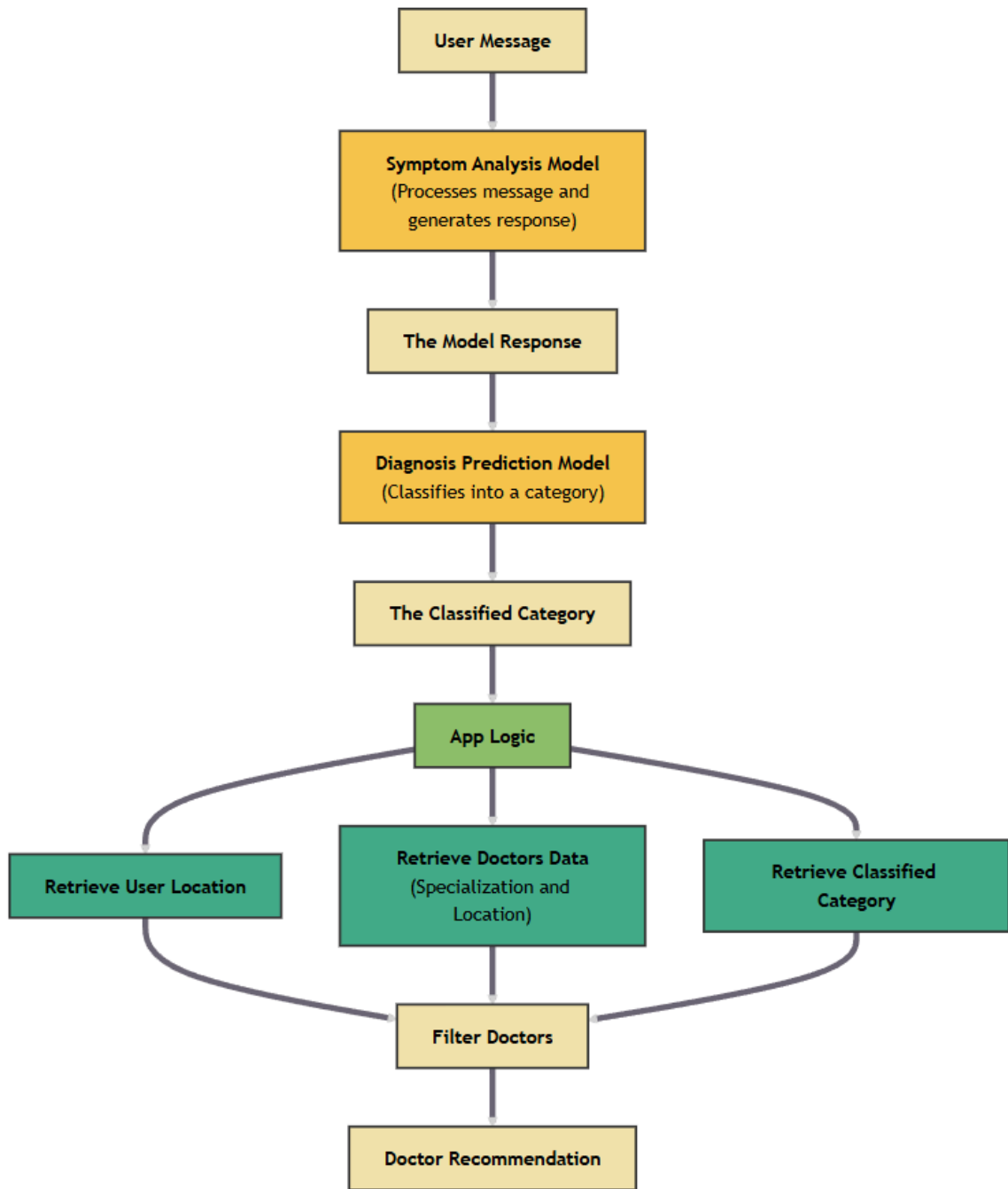


Figure 1: Symptom Analysis & Doctor Recommendation Process

3.1 Dataset

3.1.1 Arabic Healthcare Dataset (AHD) :

Arabic healthcare dataset[7] is a large text dataset consisting of more than 808K questions and answers across 90 variant categories. It makes a great input for different Arabic NLP textual tasks. The dataset is scraped from “**Altibbi**”[8] medical website and is public and free to use.

3.1.2 Dataset Description :

AHD is considered one of the largest, available and variant Arabic datasets. It provides up to 90 categories with over 808k questions and answers, ensuring a strong foundation for precise text classification. The size of this dataset makes it an appropriate corpus for applying both classical and deep learning models. The dataset adopts the questions and answers with the same annotation as it appeared in the website. The average length of the questions and answers in the dataset is 115 and 152 respectively. It's kept in a raw format with preprocessing or any type of cleaning. The following table summarizes the size of each category and it's ratio:

No	Category	Size	Ratio (%)	No	Category	Size	Ratio (%)
1	أمراض نسائية	162142	16.84%	46	أعشاب طبية	402	0.04%
2	الصحة الجنسية	96398	10.01%	47	جراحة المسالك البولية	392	0.04%
3	أمراض العضلات والعظام و المفاصل	45826	4.76%	48	الطب البديل	387	0.04%
4	أمراض المسالك البولية والتناسلية	42599	4.42%	49	العقم	386	0.04%
5	الأمراض الجلدية	41760	4.34%	50	علم النفس	382	0.04%
6	الطب العام	41740	4.34%	51	جراحة نسائية	260	0.03%
7	أمراض الجهاز الهضمي	36554	3.80%	52	تشخيص	256	0.03%
8	أمراض الأطفال	23818	2.47%	53	هرمونات	240	0.02%
9	الأمراض الجنسية	22165	2.30%	54	مختبر	224	0.02%
10	أمراض القلب والشراب	22013	2.29%	55	الامراض الوراثية	166	0.02%
11	أمراض نفسية	19091	1.98%	56	فيتامينات ومعادن	150	0.02%
12	الحمل والولادة	18763	1.95%	57	علم المناعة	92	0.01%
13	امراض العيون	18682	1.94%	58	علم الأشعة	83	0.01%
14	أفد، آذن وحنجرة	18674	1.94%	59	بصریات	82	0.01%
15	أمراض باطنية	18123	1.88%	60	خدمات طبية	77	0.01%
16	جراحة عامة	17348	1.80%	61	إسعاف أولي	68	0.01%
17	أمراض الأسنان	16901	1.75%	62	علم الأجنة	66	0.01%
18	تغذية	16344	1.70%	63	علم التشريح	61	0.01%
19	علم الصيدلة	14890	1.55%	64	علم الأمراض	57	0.01%
20	الأورام الخبيثة والحيدة	12361	1.28%	65	تخدير	56	0.01%
21	الأمراض العصبية	10845	1.13%	66	علم السموم	55	0.01%
22	الصحة النفسية	10349	1.08%	67	علم الأحياء	49	0.01%
23	أمراض العدد الصماء	9237	0.96%	68	علم الأحياء الدقيقة	48	0.00%
24	صحة الطفل	8270	0.86%	69	علم الوراثة	44	0.00%
25	مرض السكري	7147	0.74%	70	فيزيولوجيا	34	0.00%
26	البشرة والجمال	6656	0.69%	71	الطب الوقائي	34	0.00%
27	أمراض الجهاز التنفسي	6102	0.63%	72	الأمراض الرئوية	34	0.00%
28	ارتفاع ضغط الدم	4557	0.47%	73	مطاعيم و لقاحات	32	0.00%
29	أمراض الفم	4411	0.46%	74	تشوهات خلقية	21	0.00%
30	جراحة العظام	4194	0.44%	75	كيمياء	18	0.00%
31	صحة الأسنان	3967	0.41%	76	تاريخ الطب	18	0.00%
32	صحة الرجل	3806	0.40%	77	صحة المسنين	17	0.00%
33	صحة المرأة	3608	0.37%	78	جراحة الأطفال	16	0.00%
34	أمراض الدم	2362	0.25%	79	رمضان	13	0.00%
35	صحة عامة	2307	0.24%	80	كيمياء حيوية	12	0.00%
36	طب الأسنان	2232	0.23%	81	جراحة الأوعية الدموية	12	0.00%
37	إدمان	1915	0.20%	82	حمل الأنابيب	12	0.00%
38	جراحة تخمير	1579	0.16%	83	تجهيزات طبية	8	0.00%
39	الصحة والرياضة	1114	0.12%	84	غذاء	8	0.00%
40	جراحة الفك والأسنان	685	0.07%	85	أمراض الاستقلاب	7	0.00%
41	جراحة الأعصاب	593	0.06%	86	علم النبات	5	0.00%
42	جراحة القلب والشراب	575	0.06%	87	علم الأسماك	5	0.00%
43	أرجحة حساسية	555	0.06%	88	أخبار الطبي	2	0.00%
44	علاجات طبيعية	418	0.04%	89	كيمياء عضوية	1	0.00%
45	الامراض المعدية	403	0.04%	90	علم البيئة	1	0.00%

Table 1: Dataset category size and ratio before processing

3.1.3 Data Preprocessing:

The dataset is highly imbalanced, with a significant disparity in category sizes. The largest category, "أمراض نسائية" (Gynecological Diseases), contains 162,142 samples, making up a large portion of the dataset. In contrast, some categories, such as "كيمياء عضوية" (Organic Chemistry) and "علم البيئة" (Ecology), have only one sample each. This imbalance can pose challenges in machine learning tasks[9], particularly in classification, as models may become biased toward the dominant categories. Categories with over 100,000 samples (like Sexual Health with 96,398 samples) are well-represented, while those with fewer than 500 samples (like Vascular Surgery with 12 samples) might not provide enough training data for robust learning. The dataset's imbalance poses significant challenges for machine learning models, primarily leading to overfitting to majority classes, where the model becomes biased toward well-represented categories while struggling to learn from underrepresented ones. This imbalance can result in misclassification and poor recall for rare categories, as the model does not receive enough examples to

generalize effectively. Additionally, the presence of extremely small classes increases the risk of these categories being ignored altogether. To mitigate these issues, resampling techniques such as oversampling the minority classes or undersampling the dominant ones can help balance the dataset. Another approach is adjusting class weights during training, ensuring the model assigns greater importance to underrepresented categories. Without these interventions, the dataset's imbalance could negatively impact classification performance, particularly in scenarios where identifying rare cases is critical.

Merging similar categories in the dataset can significantly improve balance and model performance by reducing fragmentation [10]. For instance, "أمراض نسائية" (Gynecological Diseases) now includes related topics such as "أمراض الحمل والولادة" (Pregnancy and Childbirth) and "جراحة نسائية" (Gynecological Surgery), ensuring that all aspects of women's health are treated as one category rather than separate, underrepresented ones. Similarly, "الصحة النفسية" (Mental Health) combines categories like "أمراض نفسية" (Psychological Disorders) and "إدمان" (Addiction), capturing all mental health concerns under a broader umbrella.

By consolidating these categories, the dataset becomes more balanced, preventing underrepresented classes from being ignored while ensuring the model learns from a wider and more meaningful distribution of cases. This restructuring enhances classification accuracy and makes the dataset more effective for medical applications. The following table provides a clear overview of how similar categories have been consolidated into broader medical groups, improving dataset balance and usability.

Category	Subcategories
طبيب نفسي	أمراض نفسية، علم النفس، إدمان، الصحة النفسية
طبيب غدد صماء	أمراض الغدد الصماء، مرض السكري، هرمونات
طبيب نساء وتوليد	أمراض نسائية، الحمل والولادة، جراحة نسائية، صحة المرأة
طبيب جهاز هضمي	أمراض الجهاز الهضمي
طبيب قلب	أمراض القلب و الشرايين، ارتفاع ضغط الدم، جراحة القلب والشرايين
طبيب باطني	الطب العام، صحة عامة، تشخيص، صحة المسنين، أمراض باطنية
طبيب جلدية	الأمراض الجلدية، البشرة والجمال
طبيب عظام	أمراض العضلات والعظام والمفاصل، جراحة العظام، علاج طبيعي
طبيب أطفال	أمراض الأطفال، صحة الطفل، علم الأجنة
طبيب أمراض دم	أمراض الدم
طبيب عيون	امراض العيون
أخصائي تغذية	تغذية، فيتامينات و معادن
طبيب مسالك بولية	الصحة الجنسية، الأمراض الجنسية، صحة الرجل، أمراض المسالك البولية والتناسلية، جراحة المسالك البولية
طبيب أنف وأذن وحنجرة	أنف، أذن وحنجرة
طبيب أورام	الأورام الخبيثة والحميدة
طبيب أعصاب	الأمراض العصبية، جراحة الأعصاب
طبيب جراح	جراحة عامة، جراحة الأطفال، جراحة الأوعية الدموية
طبيب تجميل	جراحة تجميل
طبيب أسنان	أمراض الأسنان، صحة الأسنان، طب الأسنان، جراحة الفك والأسنان، أمراض الفم
طبيب صدرية	أمراض الجهاز التنفسي، الامراض المعدية

Table 2: Merged subcategories and the final merged category

By consolidating these categories, the dataset becomes more balanced, preventing underrepresented classes from being ignored while ensuring the model learns from a wider and more meaningful distribution of cases. This restructuring enhances classification accuracy and makes the dataset more effective for medical applications. After merging the categories, some still have very few examples, especially those with less than 400 rows. Removing these small categories will help the model perform better. When a category has too little data, the model struggles to learn from it and might make random or incorrect predictions. It can also get distracted by these small categories instead of focusing on the ones with enough data to learn properly. By removing them, the model becomes more accurate, trains faster, and gives more reliable results for the important categories. After eliminating these categories, we end up with the following 20 final categories with 726K questions and answers :

No	Category	Size	Percentage
1	طبيب نساء وتوليد	184773	22.78%
2	طبيب مسالك بولية	165360	20.39%
3	طبيب باطني	62443	7.70%
4	طبيب عظام	50438	6.22%
5	طبيب جلدية	48416	5.97%
6	طبيب جهاز هضمي	36554	4.51%
7	طبيب أطفال	32154	3.97%
8	طبيب نفسي	31737	3.91%
9	طبيب أسنان	28196	3.47%
10	طبيب قلب	27145	3.35%
11	طبيب عيون	18682	2.30%
12	طبيب أنف وأذن وحنجرة	18674	2.30%
13	طبيب جراح	17376	2.14%
14	طبيب غدد صماء	16624	2.05%
15	أخصائي تغذية	16494	2.03%
16	طبيب أورام	12361	1.52%
17	طبيب أعصاب	11438	1.41%
18	طبيب صدرية	6505	0.80%
19	طبيب أمراض دم	2362	0.29%
20	طبيب تجميل	1579	0.19%

Table 3: Final categories with 726K questions and answers

3.1.4 Subset dataset :

Given the large size of the dataset, training a model on the entire corpus at once is computationally expensive and time-consuming. To address this, a 5% subset (40,115 rows) was used in phase one to train the chatbot interface. Focusing on responses with answer lengths between **80–127 tokens**, ensures informative and context-rich training data for symptom analysis. For phase two we increased the subset to more than 100k rows with answer lengths between **65-350 tokens**, which results in better generalization, better performance for rare classes and higher generation.

The dataset for the classification model is made up of 9 subsets, covering 20 different medical specialties, with a total of about 108k samples. To reduce class imbalance and stay within hardware limits, we applied stratified sampling and limited each category to a maximum of 6,000 samples. During preprocessing, we cleaned the data by tokenizing the text and removing stopwords. This helped keep the input clear and meaningful before embedding it and using it to train the classification model.

3.1.5 Tokenization:

Preprocessing the dataset for NLP tasks is crucial to enhance the model's performance. Tokenization, the process of breaking text into smaller units such as words or subwords, plays a key role in structuring the text efficiently for machine learning models. Since NLP models have a maximum token limit, proper tokenization ensures that textual inputs remain within acceptable bounds. For this task, the **Natural Language Toolkit (NLTK)** was utilized, a widely adopted Python library for natural language processing. NLTK provides various tokenization tools, including word and sentence tokenizers, capable of handling different languages, including Arabic. It ensures that words are split appropriately while considering punctuation and language-specific rules.

3.1.6 Stop Word Removal:

Another essential preprocessing step is removing stop words, which are common words that do not contribute significantly to meaning, such as "في" (in) or "من" (from). Arabic stop words are particularly important to remove because they consume unnecessary token space, reducing model efficiency. Cleaning the dataset by eliminating such words helps reduce noise, improve training speed, and ensure that the model learns meaningful patterns instead of filler words[11]. This step significantly enhances the model's ability to focus on relevant linguistic structures and improves overall performance in symptom analysis.

3.2 Used Models

3.2.1 Aya Expanse 8B:

The model utilizes an optimized transformer architecture[13] in a decoder-only, auto-regressive configuration. Post-training procedures include supervised fine-tuning, preference training, and model merging. The models are optimized for multilingual tasks and support a context length of up to 8,000 tokens. Aya Expanse has been fine-

tuned using the Aya Collection, a comprehensive multilingual instruction fine-tuning dataset. This collection comprises 513 million prompts and completions across 114 languages, including Arabic.

The tokenizer used in the Aya Expans 8B[12] model is designed to handle text input across the 23 supported languages, including Arabic. While specific details about the tokenizer implementation are not provided by developers, it is tailored to effectively process and generate text with an extensive vocabulary size of more than 255,000.

3.2.2 Specialization Classifier:

We designed a hybrid deep learning architecture of convolutional and recurrent components for multi-class classification of Arabic medical text into one of 20 predefined medical specialties. Embeddings are passed through two sequential 1D convolutional layers, each using ReLU activation. The output is then processed by a bidirectional Long Short-Term Memory (BiLSTM) layer. Finally, a fully connected layer maps the learned features to 20 output classes.

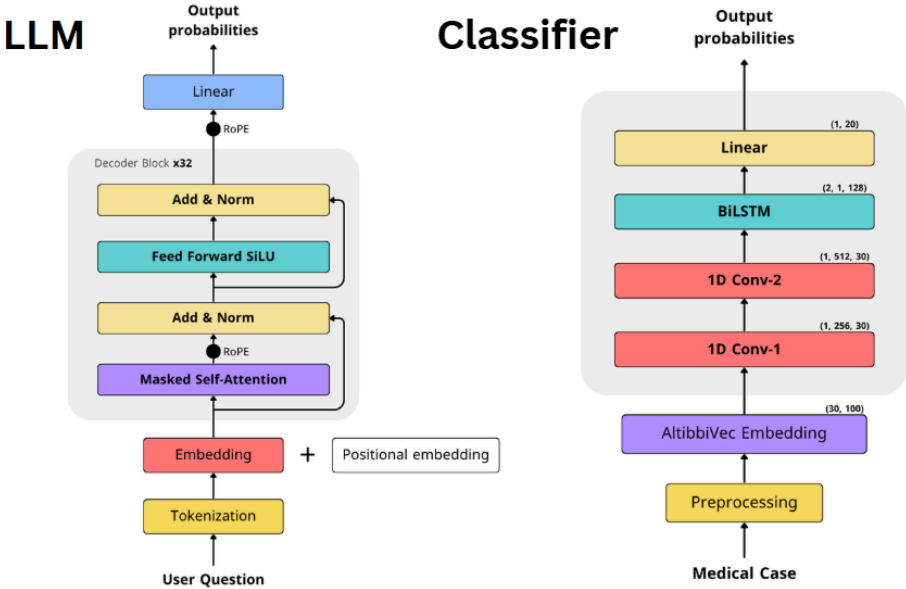


Figure 2: Aya Expans architecture on the left and classification model on the right

3.3 Doctor Database

To support personalized medical recommendations, our system incorporates a custom-built Doctors Database consisting of approximately 400 AI-generated records. These records span 20 medical specialties and are geographically restricted to Egypt. All data entries are written in Arabic to ensure consistency with the chatbot’s conversational interface and to improve language-specific relevance during retrieval.

The database was designed with user privacy in mind, using AI-generated doctor profiles instead of real personal data. It includes essential information such as medical specialty, location details, and contact fields that facilitate appropriate matching and referral.

This dataset is stored on Cloudinary, which is integrated into the backend infrastructure to enable secure and flexible access. Although Cloudinary is typically utilized for storing media content, it was adapted in our system to host structured documents and facilitate their retrieval.

The Flutter frontend interacts with this database through the backend to provide location-aware doctor recommendations. Based on the user's real-time GPS data and the predicted medical specialty from the classification model, the system suggests the most relevant nearby doctors. This connection between AI-driven symptom interpretation and physical healthcare access adds a critical layer of utility to the Dawini system.

3.4 Used Technologies

The mobile application was developed using Flutter and Firebase, where patients' and doctors' personal data are securely stored in Firebase. Generated medical reports are stored separately in a secure third-party cloud storage service provided by Cloudinary.

To efficiently train and deploy our LLM, rather than using Kaggle, we utilized a cloud-based platform optimized for large-scale AI workloads named Lightning Studio. The training and fine-tuning processes were executed on an NVIDIA L4 GPU (24 GB VRAM). After training, the final models were wrapped using FastAPI deployed through Lightning Studio's inference service with token-based authentication to protect our API from unauthorized access.

4. Experimental Work and Results

4.1 LLM Configurations:

Fine-tuning was performed on the Aya-8B model CohereForAI/aya-23-8B architecture[12], using Parameter-Efficient Fine-Tuning (PEFT) with Low-Rank Adaptation (LoRA) to minimize computational overhead. LoRA targeted projection layers including `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, and `down_proj`, with configuration parameters set to `rank=8`, `alpha=16`, and `dropout=0.1`. Training was conducted using the SFTTrainer[14] from the Transformer Reinforcement Learning (TRL) library, the setup employed an effective batch size of 32 through a combination of per-device batch size (2) and gradient accumulation steps (16). The training was conducted with a learning rate of $2e-5$, weight decay of 0.001, warmup ratio of 0.05, constant learning rate scheduling, and BF16 mixed precision. Gradient checkpointing was enabled to optimize memory usage during training.

4.2 Classifier Configurations:

The classification pipeline began with a text embedding layer utilizing AltibbiVec[4] (a domain-specific word embedding model for health and medical-related content in the Arabic context). The 100-dimensional FastText-CBOW version was used, with input sequences standardized to a length of 30 tokens, producing tensors of shape (batch_size, 30, 100). The neural architecture comprised a hybrid CNN-BiLSTM model: the convolutional component included two Conv1D layers (100→256 and 256→512 channels, respectively) with kernel size 3 and

ReLU activation, followed by a bidirectional LSTM layer with 1 layer, 512 input features, and 128 hidden units. The classification head consisted of a fully connected layer mapping the LSTM outputs to the 20 medical categories. Training was performed using the Adam optimizer with a learning rate of 1×10^{-3} and cross-entropy loss, with a batch size of 64 over 5 epochs.

A series of experiments conducted to optimize the medical specialty classification model. The experiments varied in terms of dataset size, model architecture, the following table summarizes the results:

#	Embedding Dim	Samples	Model Architecture	LR	Train Loss	Train Acc	Val Acc	Precision	Recall
1	100	2000	1 Conv, 1 LSTM, 1 FC	1e-3	0.9646	0.70	0.65	0.65	0.64
2	100	3000	1 Conv, 1 LSTM, 1 FC	1e-3	0.8116	0.75	0.74	0.74	0.74
3	100	5000	1 Conv, 1 LSTM, 1 FC	1e-3	0.7668	0.76	0.74	0.74	0.74
4	100	5000	2 Conv, 1 LSTM, 2 FC	1e-4	0.6065	0.81	0.77	0.76	0.77
5	100	6000	2 Conv, 1 BiLSTM, 1 FC	1e-3	0.5363	0.83	0.77	0.77	0.78
6	100	6000	2 Conv, 5 BiLSTM, 1 FC	1e-3	0.6966	0.79	0.76	0.76	0.76

Table 4: Results of classification experiments with different architectures and datasets

As shown in the table, increasing the dataset size and refining the model architecture led to consistent improvements in performance, with the best results achieved using larger datasets and deeper architecture, combining two convolutional layers followed by one BiLSTM layer.

4.3 Evaluation & Results:

The model was evaluated on 130 validation samples. Text generation was controlled using parameters: temperature of 0.8, top-p of 0.8, top-k of 40, a maximum of 512 new tokens, and sampling enabled, while cosine similarity based on TF-IDF vectorization and semantic match were used, the model achieved a cosine similarity score of 0.22 and semantic match score of 0.55. We also monitored the mean token accuracy in each batch to make sure the model did not overfit on training data, the result was 69.71%, 64.66%, and 71.43% for stage 1, 2 and 3 respectively.

In the loss graph, we can see a clear improvement across stages, but each stage transition introduces temporary destabilization in the loss due to dataset differences, however, we should consider that each dataset shift caused a loss spike, which suggest the datasets were meaningfully different.

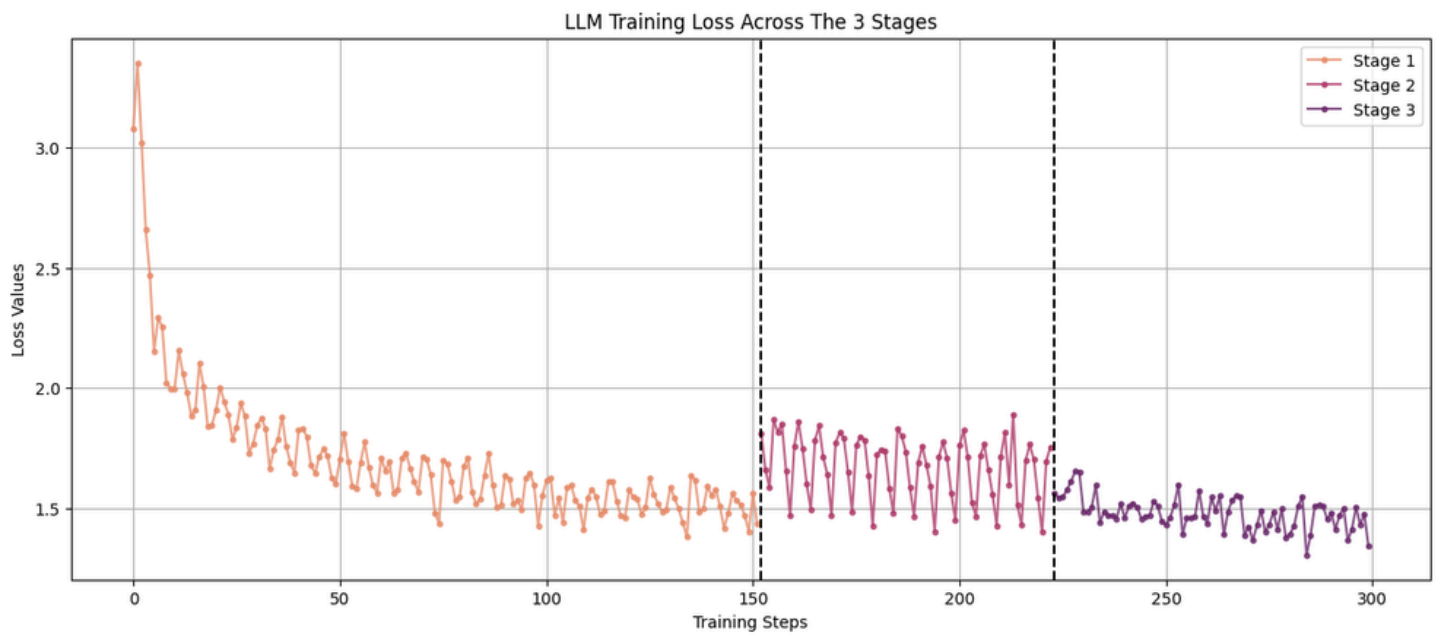


Figure 3: Loss Graph of Aya Expanse during STF process across the 3 fine-tuning stages

Also, the increase of loss in stage-2 likely reflects a change in data characteristics, since it contains answers of more variable lengths (shorter and much longer answers).

Question	Doctor Answer	LLM Answer	Original Category	Predicted Category
وقعت على سن السلم و الخبطه جت تحت ركبتى مباشره و عملتلى كدمه و تورم و الخبطه بقالها 10 ايام و التورم لسه مخفش هل التورم بياخذ وقت طويل	يزول التورم تحت الركبه بعد اصابه الركبه والقدم من تلقاء نفسه، ولكن قد يستغرق ذلك عده ايام او حتى اسابيع. هناك بعض التدابير التي يمكنك القيام بها لمساعدته التورم على الشفاء بشكل اسرع، مثل: تجنب النشاط الذي يسبب الالم او التورم، وضع الثلج على المنطقه المصابه لمدته 20 دقيقه كل ساعه لمدته 24 ساعه الاولى.	نعم ياخذ وقت طويل جدا ، لكن في الوقت الحالي يجب عمل اشعه للركبه للتأكد من عدم وجود كسر او اصابه في الاربطه او غضاريف الركبه ، بعد التأكد من عدم وجود كسر او اصابه ، يمكن البدء في العلاج الطبيعي للركبه	طبيب عظام	طبيب عظام
ابني عنده 16 سنه تركيزه ضعيف جدا سريع النسيان حتي في الدراسه عكس ماكان عليه في الطفوله فما هو الحل ومن هو الطبيب المتخصص في علاجه	السلام عليكم المراهقه مرحله مهمه للشخص ذكر انا اثني فيها ينطج جسديا وعقليا ومن المهم التعامل معها بجديه وعدم اغفال التفاصيل ففيها تنشيط الهرمونات وقد تسبب اعراض كثيره من تشتت ذهني الى العصبيه الى التوتر العاطفي وقد تزيد من شدة الحزن والانفعال لدى المراهق ومن المهم الوقوف الى جانب المراهق في هذه الفتره وعدم	السلام عليكم ورحمه الله وبركاته اولا يجب التأكد من عدم وجود اي مشكله صحيه او عضويه تؤثر على التركيز مثل مشاكل في السمع او الرؤيه او نقص في هرمونات الغده الدرقيه او الهرمونات الجنسيه ويجب عمل فحص دم شامل ثانيا في حاله نفي وجود اي مشكله عضويه او صحيه يجب استشاره طبيب نفسي للكشف عن اي مشكله نفسيه	طبيب باطني	طبيب أعصاب

		او عصبية مثل القلق او الاكتئاب او التشنجات او اضطراب الانتباه والفرط الحركة ثالثا في حاله نفي وجود اي مشكله عضويه او صحيه او نفسيه يجب استشاره طبيب اطفال لوصف العلاج المناسب		تركه نصيحتي هي الابتعاد عن ايه طب شعبي او عقاقير تسمعون عنها ومراجعه طبيب نفسي مختص طبعا بعد اقناع ابنكم بان هذا الاجراء في صالحه والطبيب النفسي قد يطلب منكم مجموعه من التحليل للتأكد بشكل كامل بالاضافه الى الفحص , ف , العاده
طبيب غدد صماء	طبيب باطني	هذا ارتفاع بسيط لا يدل علي الاصابه بالسكري ولكن يجب المتابعه مع طبيب مختص لمتابعه السكر لان هذا الارتفاع قد يكون ناجما عن مشاكل في الغده الدقيه او اضطرابات اخري في الجسم لذلك يجب عمل فحص شامل للسكر بعد ٨ ساعات من الاكل واخذ عينه من الدم واختبارها علي جهاز قياس السكر في الدم لذا يجب المتابعه مع طبيب مختص للكشف عن اي اضطراب في الجسم قد يكون السبب في ارتفاع السكر	سلامتك,,, الوقت اللازم للصيام حتى تكون قراءه السكري المعتمده على الصيام دقيقه هي 8 ساعات على الاقل : حيث تظهر قراءات فحصك للسكري انها طبيعيه , اضافه الى ان قراءه السكري التراكمي ايضا هي طبيعيه (الطبيعي من 4 - 6) , لكن سبب الارتفاع البسيط للسكري عند الفحص بعد 8 ساعات هو بسبب القلق والتوتر المصاحب للفحص , حيث انه عند القلق والتوتر ترتفع هرمونات معينه في الجسم تؤدي بالنهايه الى رفع السكري بشكل مؤقت , كما حدث عندك . لذلك اطمئن عزيزي ,	السكر بعد صيام 6 ساعات اجده ما بين ال100و112ولكن عند الصوم ل 8 ساعات اجده 99..104..85..96 ل ارتفاع السكر الصومي هذا يدل علي مقدمات السكري او الاصابه به

فوضعك الصحي
مطمئن ، وهدئ
من روعك وقلل
القلق

Table 5: Some text generation and question classification examples from the validation dataset

The LLM was evaluated on a set of clinical scenarios covering different specialties. As shown in the table, the model generated medically coherent answers that addressed the patient's concerns and included appropriate diagnostic or therapeutic advice. In terms of clinical reasoning, the model often expanded on the human doctor responses, Also it can make a significant mistakes, as in case 3 in the table, but we prompted it to add extra safety by stating that it could make mistakes.

Classification model evaluated on the validation set employed the standard classification metrics. The final validation accuracy was 77.35%, with a mean precision of 0.7792 and mean recall of 0.7824.

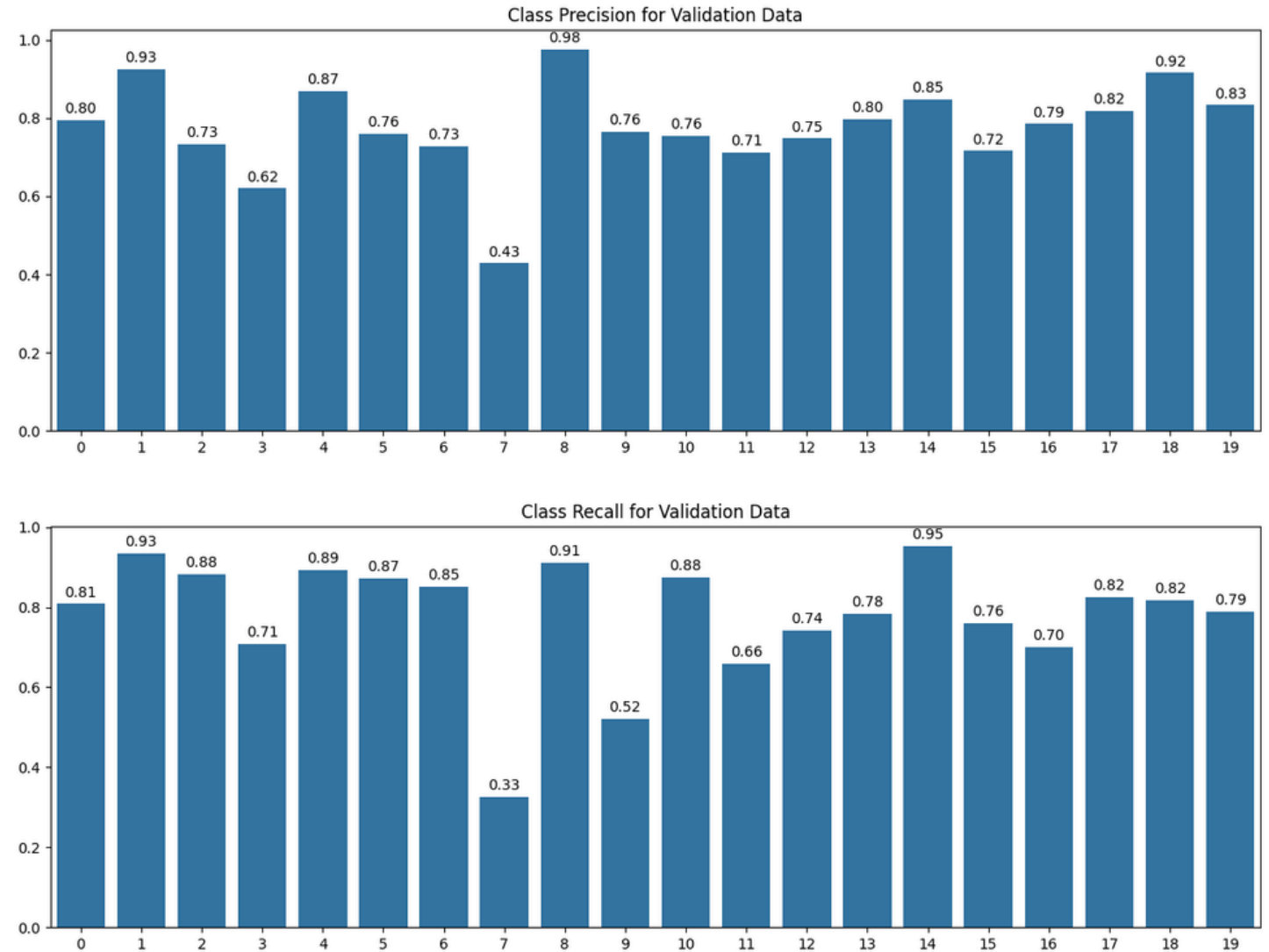


Figure 4: Precision and recall across the 20 classes from validation dataset

For medical specialty prediction, the model correctly identified the target specialty in one out of three cases. The two misclassified cases involved domain overlaps where the model leaned toward specialties more aligned with the diagnostic approach mentioned in its response rather than the minimal specialty

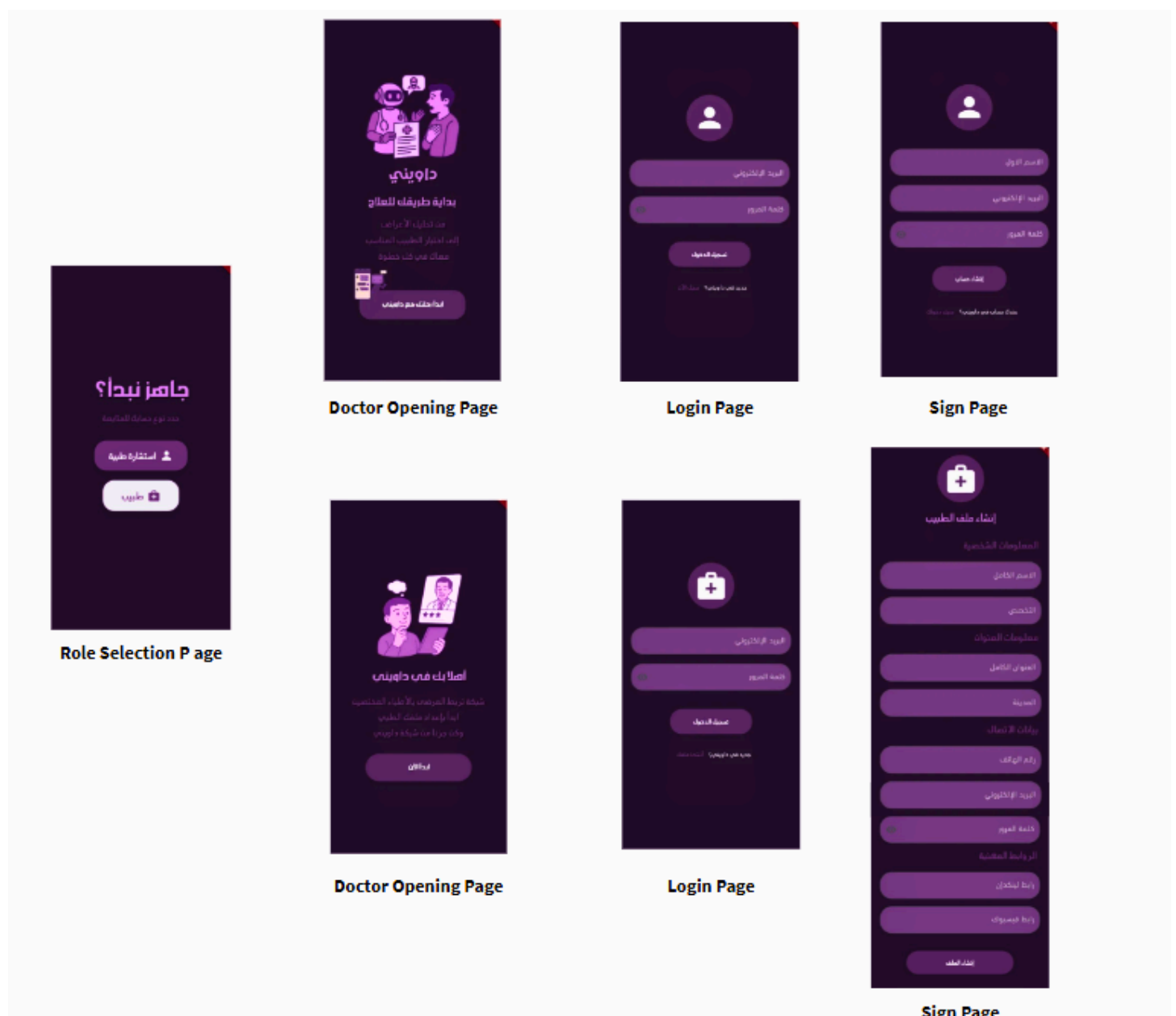
suggested in the original answer. For example, the second case in the table was attributed to a neurologist rather than an internist, possibly due to semantic overlap between neurological and psychological symptomatology in the training data.

5. User Interface

5.1 Application Overview

The mobile application developed for this graduation project is a healthcare platform that facilitates communication between patients and healthcare professionals. The system is designed to support two distinct roles: **User (Patient)** and **Doctor**, with each role offering a customized interface and flow to suit its functional requirements.

Upon launching the application, users are directed to the authentication flow, which is implemented using **Firestore Authentication**. This service provides secure user registration and login using email and password credentials. Users are prompted to choose their role—either as a user or a doctor. This selection determines the navigation flow, data access, and available features within the application, ensuring that each user only interacts with components relevant to their role.



5.2 User Role (Patient)

Once a user logs into the application and selects the **User (Patient)** role, they are directed immediately to the main **Chat View** interface. This screen allows the user to communicate directly with an AI-powered model integrated into the system. The user can engage in real-time conversation through a text input field and receive intelligent responses from the model. Each chat is stored securely using **Firebase Firestore**, ensuring that all chat sessions are saved and can be accessed later through the **Chat History** side drawer. The user also has the option to create a new chat or delete previous ones as needed, providing a flexible and user-controlled messaging experience.

In addition to chat functionality, the user can access their **profile page**, where they can update basic information such as their name, email address, and password. The user can also choose to log out or permanently delete their account. As this functionality is common in most mobile applications, it is implemented with standard UI/UX conventions and does not constitute a core feature of the system.

One of the central features for users is the ability to receive **doctor recommendations**. Below the chat input field, there is a dedicated button labeled "Recommend Doctors." Upon clicking this button, the user is navigated to a **Medical Form** page, where they are prompted to fill in personal and health-related information. This includes data such as height, weight, and other general details, as well as a critical section called **Case Details**, where the user is asked to describe their symptoms and provide relevant medical context.

When the form is submitted, two backend models are called:

1. **Specialization Classifier Model:** This model receives the *case details only* and is responsible for predicting the appropriate **medical specialization** (e.g., dentistry, neurology, general medicine) required for the case. The result determines which group of doctors the user will be shown.
2. **Aya Expanse 8B Model:** The second model processes the full form, including both personal and medical details, and generates a **medical report**. This report is saved as a PDF file and uploaded to **Cloudinary**, after which its URL is stored in the user's Firestore document for future reference.

Once processing is complete, the user is navigated to the **Recommended Doctors** page, which displays a list of doctors belonging to the predicted specialization. On the user's first visit to this screen, they are prompted once to grant **location access**. If permission is granted, the system retrieves the user's geographic coordinates and calculates the distance between the user and each doctor on the list.

The page also includes a **location-based filter button** located in the corner of the screen. When this button is activated, it filters the list to show only doctors within a **30-kilometer radius** of the user's current location. Clicking the button again restores the full list. This helps users find doctors who are more accessible based on their proximity.

By tapping on any doctor from the list, the user is taken to the **Doctor Profile** page. This screen displays the doctor's name, specialty, description, and approximate location. The user can also click a button labeled "Send Report," which sends the previously generated medical report to the selected doctor, allowing them to review the patient's case.

Additionally, next to the displayed address, there is a small button that opens **Google Maps**, pinpointing the exact location of the doctor's clinic or office. This ensures that users can easily navigate to the physical location if needed.

The user role is designed to offer an intelligent, interactive experience that combines AI-driven medical insights with traditional healthcare communication tools. From chatting with the AI model to receiving personalized doctor recommendations based on symptoms, the user flow is streamlined and supported by a backend infrastructure leveraging Firebase, AI models, and cloud-based services.

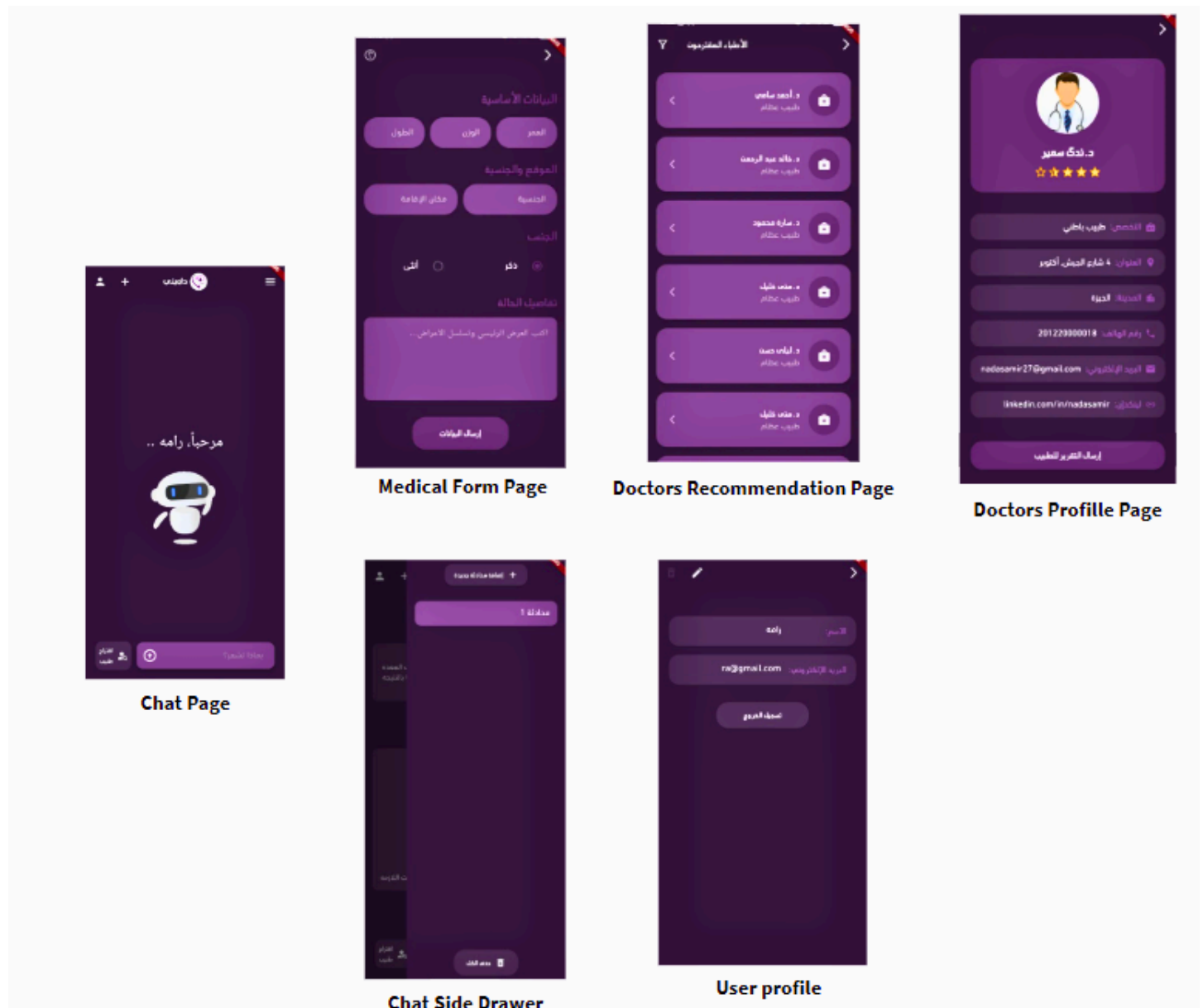


Figure 6: User Role (Patient)

5.3 Doctor Role

Doctors who log into the application are directed to their **Profile Page**, where they can view and manage their personal and professional details. This includes the ability to update information such as their name, specialty, and description. Additionally, doctors have the option to **delete their account** if they choose to discontinue their use of the application.

A key feature available to doctors is access to **patient reports**. When a patient sends a report, it is stored and linked to the respective doctor's account. Within the profile page, doctors can navigate to a dedicated **Reports Page** that displays a list of all reports submitted to them by users. These reports are presented in an organized format to allow doctors to easily browse and review them.

While the content and structure of these reports are discussed in detail later in this report, it is important to note that they serve as a valuable resource for doctors to understand the patient's case before proceeding with further actions.

This role-specific flow is designed to keep the doctor's experience focused and efficient, allowing them to manage their professional presence in the app and stay informed about their patients in a streamlined and secure manner.

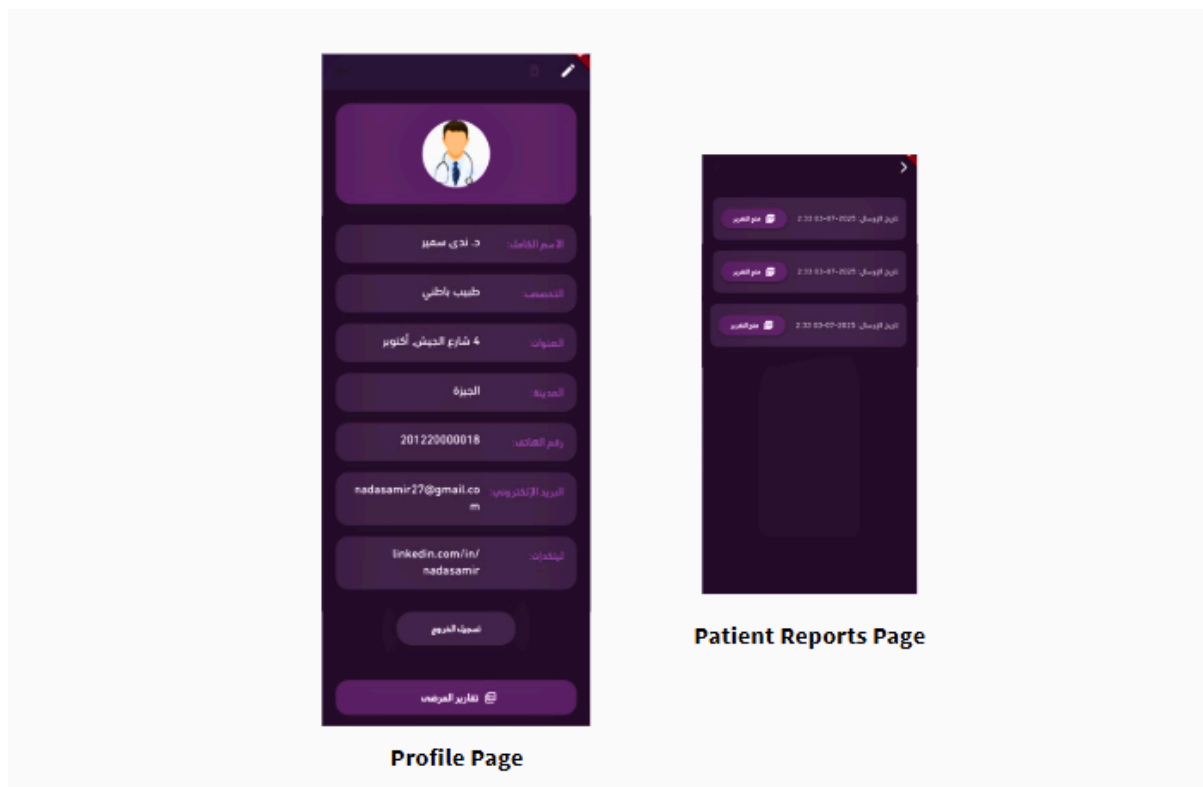


Figure 7: Doctor Role

As part of the system's scope, the application also considers supporting more dynamic interactions between users and doctors. This includes the ability for users to initiate direct conversations with doctors and optionally request appointments when needed. Additionally, there is room to expand the doctor's profile with more detailed information and structured data, allowing for a richer and more informative experience while preserving the app's simplicity and clarity.

Conclusion

Dawini is not just a chatbot—it is a step toward bridging the gap between healthcare and Arabic-speaking communities. Our vision was to create a reliable digital health companion that helps everyone understand their symptoms and receive guidance in their own language. By combining Arabic Natural Language Processing with medical expertise, Dawini delivers both clinically relevant answers and specialty recommendations tailored to user needs.

To make this possible, we fine-tuned a large Arabic language model using domain-specific data. Dawini achieved a similarity score of 0.22 and a semantic match score of 0.55 on the text generation task. For classification, it reached a cosine similarity of 0.22, semantic match of 0.55, and 77.35% accuracy in predicting the correct medical specialty. These results show the system's ability to generate informative and

safety-focused responses, while also guiding users to the right type of care—although some confusion remains in overlapping specialties.

We invested time in preparing high-quality datasets and refining the system to understand real-world patient input. Our goal was to make medical help more accessible, personalized, and culturally relevant, especially in regions where Arabic healthcare tools are limited.

In the future, we aim to improve Dawini's classification accuracy, expand the dataset to include more diverse and dialectal inputs, and collaborate with healthcare providers and platforms to bring this tool closer to real clinical environments. Ultimately, Dawini represents a major step toward making healthcare support available anytime, anywhere—in your pocket, in your language.

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